

From Observations to Climate Services: The Journey of One observation

Every observation tells a story. Some begin in a ship logbook, others in a handwritten station register, while many are recorded automatically by modern observing systems. Yet all of them share the same journey before they can contribute to climate services.

Observations are the foundation of weather forecasting, climate monitoring, reanalysis, climate prediction and climate services. However, their value is not determined solely by their existence. It depends on a journey involving preservation, quality assessment, stewardship, data management, international exchange and transformation into climate information. For historical observations, this journey begins even earlier with archives, data rescue and digitization.

This presentation follows the life cycle of a single observation—from its original recording to its preservation, quality control, metadata documentation, interoperability and international exchange, before becoming part of climate datasets, monitoring products, reanalyses and climate services supporting decision-making. Drawing on recent WMO initiatives and international collaborations, it provides an integrated vision of how many communities contribute to one shared journey, transforming observations into trusted climate information for society.